NRLC-02: NRQL → DQL Translation

> **Series:** NRLC — New Relic to Dynatrace Migration Deep Dives | **Notebook:** 2 of 9 | **Created:** April 2026 | **Last Updated:** 08/27/2026

Overview

Query translation is the technical core of any New Relic → Dynatrace migration. NRQL is SQL-like and declarative; DQL is pipeline-based. A direct string substitution doesn't work — the structure, source resolution, and time semantics all differ. This deep dive covers the **architecture and behavior of the open-source `nrql-engine` compiler** (292 tested patterns, AST-based), the translation surface it covers, the confidence scoring it produces, the gaps it flags for manual review, and the patterns to recognize when reading its output.

The TypeScript `nrql-engine` and the Python compiler in `Dynatrace-NewRelic/compiler/` are **pinned to each other via the Phase 19b regression suite** (`test_phase19b_engine_parity.py` in CI) — drift on either side trips the test. Both expose the same 292-pattern surface with identical confidence scoring.

Table of Contents

11. Validation Loop

Prerequisites

Requirement	Details
Audience	Engineers translating NRQL queries; reviewers of automated translations
Recommended	NRLC-01 (platform comparison); familiarity with both NRQL and DQL syntax
Companion projects	`nrql-engine`, `nrql-translator`, `NewRelic-to-Dynatrace-Migration-Utilities`

Engine Support (Phase 19b)

The compiler is exposed through three projects (all pinned to the same 292-pattern surface via Phase 19b):

Surface	Module / CLI
Single NRQL query	`migrate.py compile "<nrql>"` or `nrql-translator query`
Batch translation	`migrate.py batch --file queries.csv` or `nrql-translator excel`
Translate + auto-fix + post-processing	`migrate.py convert --file queries.txt` (wraps compile + DQLFixer + Phase 19 apdex/compare/rate/percentage uplift)
Validate syntax only	`migrate.py compile --validate`
Shorthand pre-expand (`Distributed*`, `Mobile*`, `Lambda*`, etc.)	`compiler/shorthands.py` (9 patterns; Phase 19b)

Surface	Module / CLI
DQL auto-fix rules	`validators/dql_fixer.py` (24 methods; Phase 19b one-to-one with TS `dql-fixer.ts`)

See COVERAGE-MATRIX.md §1 APM for the full NRQL event-class → DT data-object mapping (Transaction, Span, Log, Metric, SystemSample, PageView, MobileSession, SyntheticCheck, and custom event types).

1. Why a Compiler, Not Regex

Early translation tools used regular expressions: find `SELECT count(*)`, replace with `summarize count = count()`. This breaks fast:

Failure mode	Why regex fails
Nested function calls	`percentage(count(*), WHERE status = 200)` requires recursion, not pattern match
`FACET CASES` with nested `if()`	Requires understanding the AST to emit nested DQL
Subqueries / `SELECT ... FROM (SELECT ...)`	Regex can't track scope
Time expressions	`SINCE 1 day ago UNTIL 1 hour ago` vs. `SINCE bin(now(), 24h)` need semantic awareness
Compound `WHERE` with mixed operators	Operator precedence requires a parser

`nrql-engine` replaced an early 2,197-line regex translator with a proper compiler. The result: 292 tested translation patterns, **1,200+ unit tests in the Python orchestrator (post-Phase-24)**, and the ability to translate complex queries that the regex tool silently produced wrong DQL for. Phase 19b added a CI parity job (`test_phase19b_engine_parity.py`) that pins the Python compiler to the TS `nrql-engine` — drift on either side trips the test.

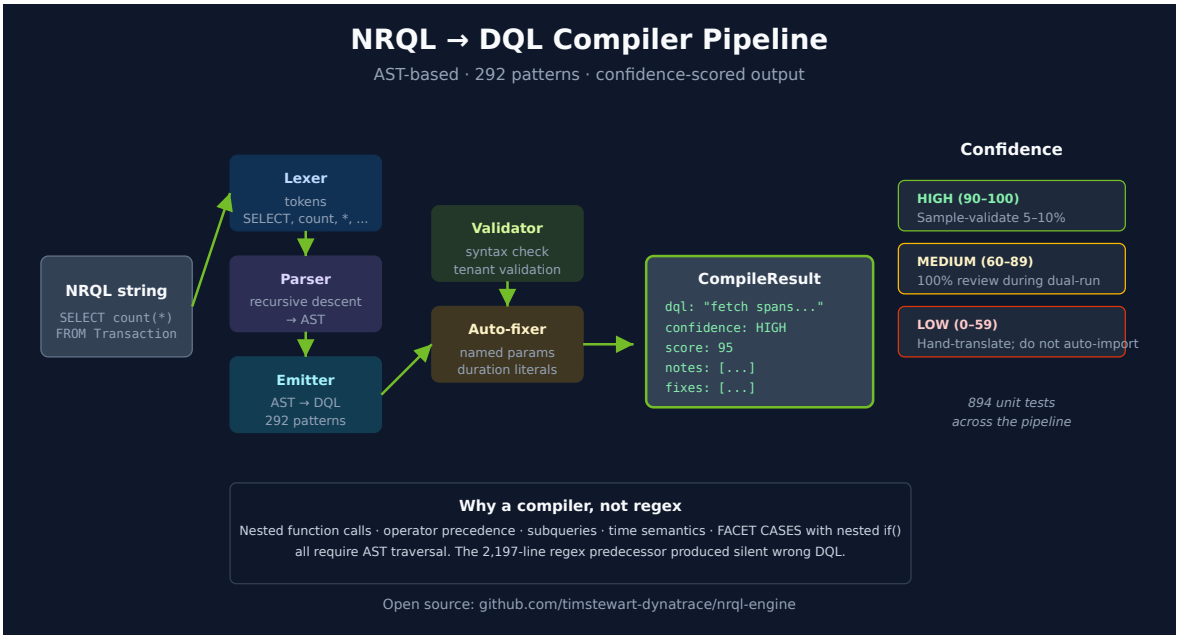
2. Compiler Pipeline

```
NRQL string
    ↓
  Lexer       → token stream
    ↓
  Parser      → AST (recursive descent)
    ↓
  Emitter     → DQL string + CompileResult
    ↓
  Validator   → syntactically valid? against tenant?
    ↓
  Auto-fixer  → attempt safe corrections (operator normalization, duration
literals)
    ↓
Output: { dql, confidence, notes[], warnings[], fixes[] }
```

Each stage lives in a separate module: `compiler/lexer.py`, `compiler/parser.py`, `compiler/ast_nodes.py`, `compiler/emitter.py` (Python project), or `src/compiler/*.ts` (TypeScript engine).

The `CompileResult` is the contract every consumer uses:

```
interface CompileResult {
  success: boolean;
  dql: string;
  confidence: 'HIGH' | 'MEDIUM' | 'LOW';
  confidenceScore: number;     // 0-100
  ast?: NRQLNode;
  notes: TranslationNotes;     // structured guidance
  warnings: string[];
  fixes: string[];             // applied auto-fixes
}
```



3. Translation Surface

Supported NRQL clauses: `SELECT`, `FROM`, `WHERE`, `FACET`, `TIMESERIES`, `LIMIT`, `ORDER BY`, `COMPARE WITH`, `SLIDE BY`, `SINCE`, `UNTIL`

Supported aggregations: `count`, `average` / `avg`, `sum`, `min`, `max`, `uniqueCount`, `percentile`, `latest`, `earliest`, `uniques`, `median`, `stddev`, `rate`, `percentage`, `cdfPercentage`, `filter`, `histogram`, `funnel`, `apdex` (with caveats)

Supported expressions:

- Arithmetic between aggregations (`sum(a) / count()`)
- Nested `if()` for conditional faceting
- Time grouping (`hourOf`, `dateOf`, `weekOf`)
- Subqueries via `lookup` patterns
- `WHERE` with `AND` / `OR` / `NOT` / `IN` / `LIKE` / regex

Partial / produces warnings:

- `apdex()` — emits DQL with TODO marker; threshold needs config
- `bytecountestimate()` — no DQL equivalent; recommend reformulation
- `funnel()` — multi-stage; emitted as nested append/join with manual review note

Not supported:

- `cohort()` — retention analysis is a different DT capability (Dynatrace Intelligence or custom DQL)

- Custom event types not present in DT (manual mapping required)
- Some metric streaming integrations — routed through OneAgent or OTLP instead

4. Confidence Scoring

Every translation gets a confidence label and a 0–100 score. The scoring is rule-based, not heuristic:

Confidence	Score	Meaning
HIGH	90–100	Pattern fully covered; structural and semantic equivalence
MEDIUM	60–89	Translation correct in common case; edge cases (NULLs, empty sets, time boundaries) need spot-check
LOW	0–59	Manual review required; one or more constructs lack a direct equivalent

What lowers confidence

Construct	Confidence Impact
Use of `apdex()`	LOW (threshold not portable)
Use of `funnel()`	LOW (multi-stage; manual review)
Subquery with `lookup`	MEDIUM (correctness depends on lookup data)
`SLIDE BY`	MEDIUM (DT supports but rarely-used path)
Nested `if()` in `FACET CASES`	MEDIUM (emits nested DQL; review structure)
`COMPARE WITH`	MEDIUM (emits append + time-shift; verify alignment)
Standard `SELECT count(*) FROM X WHERE Y FACET Z TIMESERIES`	HIGH

Acting on Confidence

- **HIGH** → trust the translation; sample-validate 5–10% during dual-run
- **MEDIUM** → 100% review during dual-run; verify edge cases
- **LOW** → hand-translate or reformulate the requirement; do not auto-import

5. Pattern Library — Worked Examples

Simple aggregation with FACET

NRQL:

```
SELECT count(*), average(duration) FROM Transaction
WHERE appName = 'checkout' SINCE 1 hour ago FACET host
```

DQL:

```
fetch spans, from:-1h
| filter service.name == "checkout"
| summarize count = count(), average_duration = avg(duration), by:{host.name}
```

Confidence: **HIGH (100)**

Percentage with conditional

NRQL:

```
SELECT percentage(count(*), WHERE status = 200) FROM Transaction
```

DQL:

```
fetch spans, from:-1h
| summarize success_pct = 100.0 * countIf(status == 200) / count()
```

Confidence: **HIGH (95)** — produces the same numeric result; column name differs.

Rate (decomposed)

NRQL:

```
SELECT rate(count(*), 1 minute) FROM Transaction
```

DQL:

```
timeseries count = count(), interval:1m, from:-1h
```

Confidence: **HIGH (90)** — NRQL `rate(count, N)` is structurally a per-interval count, which DQL expresses as a `timeseries`.

COMPARE WITH (time-shifted append)

NRQL:

```
SELECT count(*) FROM Transaction COMPARE WITH 1 week ago SINCE 1 day ago
```

DQL:

```
fetch spans, from:-1d
| summarize current_count = count()
| append [
    fetch spans, from:-192h, to:-168h
    | summarize previous_count = count()
  ]
```

Confidence: **MEDIUM (75)** — verify the time alignment; DT default behavior on `append` may need a join for visualization.

Subquery via lookup

NRQL:

```
SELECT count(*) FROM PageView WHERE userAgent IN (SELECT name FROM BotList)
```

DQL:

```
fetch logs, from:-1h
| filter dt.entity.application_method == "PageView"
| lookup [fetch logs, from:-1h | filter dt.entity.application_method ==
"BotList" | fields name],
    sourceField: userAgent, lookupField: name
| filter isNotNull(name)
| summarize count()
```

Confidence: **MEDIUM (70)** — correctness depends on whether the bot list exists as a lookupable Grail data source.

6. Source-Class Mapping (FROM clause)

NRQL's `FROM` references NR event classes; DQL's `fetch` references Grail data objects.

NRQL FROM	DQL fetch	Notes
`Transaction`	`fetch spans` (filter `span.kind == "server"`)	APM transactions become server spans
`TransactionError`	`fetch spans` + `filter isNotNull(error)`	Error spans
`Span`	`fetch spans`	Direct
`Log`	`fetch logs`	Direct
`Metric`	`timeseries <agg>(<metric.key>)`	NRQL metrics → DT metrics
`SystemSample`	`timeseries avg(dt.host.cpu.usage)` (etc.)	Infrastructure host metrics
`ProcessSample`	`timeseries ... by: {dt.entity.process_group_instance}`	Process metrics
`K8sContainerSample`	`timeseries ... by: {k8s.container.name}`	K8s container metrics
`PageView`	`fetch logs` (RUM page views) or `fetch events`	Depends on RUM ingest path
`BrowserInteraction`	`fetch events` (filter rum events)	RUM-specific

NRQL FROM	DQL fetch	Notes
`MobileSession`	`fetch events` (filter mobile rum)	Mobile RUM
`SyntheticCheck`	`fetch events` (filter synthetic.* events)	Synthetic results
Custom event type	`fetch bizevents` or `fetch events`	Depends on ingest mapping

The compiler picks the right `fetch` source based on a built-in lookup; you can override via the `--source-hint` CLI flag if your NR custom event has a non-default DT mapping.

7. Function Mapping (Aggregations)

NRQL	DQL	Notes
`count(*)`	`count()`	
`count(field)`	`count(field)`	counts non-null
`uniqueCount(field)`	`countDistinct(field)`	
`average(field)` / `avg(field)`	`avg(field)`	
`sum(field)`	`sum(field)`	
`min(field)` / `max(field)`	`min(field)` / `max(field)`	
`percentile(field, 95)`	`percentile(field, 95)`	DQL uses positional; NRQL uses positional
`median(field)`	`median(field)`	
`stddev(field)`	`stddev(field)`	
`latest(field)`	`takeLast(field)`	with `sort timestamp desc` if needed
`earliest(field)`	`takeFirst(field)`	
`uniques(field)`	`collectDistinct(field)`	

NRQL	DQL	Notes
`rate(count(*), 1 minute)`	`timeseries count(), interval:1m`	structural rewrite
`percentage(count(*), WHERE x)`	`100.0 * countIf(x) / count()`	
`filter(count(*), WHERE x)`	`countIf(x)`	
`histogram(field, ...)`	`summarize ..., by: {bin(field, width:N)}`	bucket-based
`apdex(field, t:N)`	flagged TODO; manual config	DT uses Apdex via Dynatrace Intelligence or custom DQL
`funnel(...)`	flagged; nested append/join skeleton	manual review
`cohort(...)`	unsupported	use Dynatrace Intelligence or custom DQL

8. Time Expression Mapping

NRQL	DQL
`SINCE 1 hour ago`	`from:-1h`
`SINCE 1 day ago UNTIL 1 hour ago`	`from:-1d, to:-1h`
`SINCE '2026-04-14 00:00:00'`	`from:"2026-04-14T00:00:00Z"`
`TIMESERIES 5 minutes`	`interval:5m`
`TIMESERIES MAX`	`interval: auto`
`SLIDE BY 1 minute`	not directly supported — use `bin(timestamp, 1m)` for similar shape

NRQL	DQL
`COMPARE WITH 1 week ago`	second `fetch` with `from:-2w, to:-1w` plus shift

The compiler normalizes all time expressions to absolute or relative DQL forms. A common gotcha: NRQL's default time window (1 hour) differs from DQL's defaults; the compiler always emits an explicit `from:` to avoid this.

9. Unsupported NRQL — What to Do

When the compiler emits LOW confidence or a TODO marker:

Symptom	Cause	Action
Translation contains `// TODO: APDEX_THRESHOLD`	NRQL `apdex(field, t:0.5)` — threshold doesn't translate	Configure DT Apdex calculation separately or use Dynatrace Intelligence SLO with target value
Translation contains `// TODO: FUNNEL_STAGES`	Multi-stage funnel	Emit per-stage metrics + custom DQL with `lookup` chain
Compiler returns `success: false`	Unparseable NRQL (likely malformed or uses a private NR feature)	Hand-translate; capture in gap-analysis
LOW confidence on subquery	`lookup` target not validated against Grail	Check that the looked-up data exists as queryable Grail content
LOW confidence on custom event type	Source mapping unknown	Add a source mapping override or route the data via OpenPipeline first

Document gaps in the migration's gap-analysis artifact (NR2DT-01 Discover §6). LOW-confidence queries are a planned cost, not a surprise.

10. Tooling: Engine, Translator, CLI

Three projects expose the same compiler in different shapes:

Tool	Form	Use Case
`nrql-engine`	TypeScript library (`npm install @timstewart-dynatrace/nrql-engine`)	Embed in your own tool, web app, or CI — **planned future home:** `dynatrace-dma` **(Dynatrace Migration Assistant)**
`nrql-translator`	Thin CLI around the engine	Single query, batch Excel, validation, notebook generation
`NewRelic-to-Dynatrace-Migration-Utilities`	Python CLI orchestrating full migration	End-to-end NR→DT including all entity transformers

Single-query CLI

```
# Using nrql-translator
nrql-translator query "SELECT count(*) FROM Transaction FACET host"

# Or Dynatrace-NewRelic Python CLI
python3 migrate.py compile "SELECT count(*) FROM Transaction FACET host"
```

Batch translation

```
# CSV in, CSV out (with confidence column)
python3 migrate.py batch --file all-queries.csv

# Excel batch (preserves dashboard mapping)
nrql-translator excel --in dashboards.xlsx --out dashboards-dql.xlsx
```

Notebook output

The translator can emit a Dynatrace notebook directly from a batch:

```
nrql-translator notebook --in dashboard-queries.csv --out migrated-dashboard.ipynb
```

Dynatrace Migration Assistant (`dynatrace-dma`) Home

The `dynatrace-dma` GitHub organization is the upstream **Dynatrace Migration Assistant** home for source-platform migration tooling. It already hosts:

- `splunk-to-dynatrace` — Splunk → Dynatrace migration resources
- `datadog-to-dynatrace` — Datadog configuration extraction scripts

The NRQL engine is planned to relocate here under a `newrelic-to-dynatrace` repo following the same pattern. Until the move, the engine remains canonical at `timstewart-dynatrace/nrql-engine`; track `dynatrace-dma` for the official Dynatrace-supported version going forward.

11. Validation Loop

Translation alone doesn't guarantee correctness. The full loop is:

```
NRQL → [compile] → DQL → [syntax validate] → [tenant validate] → [behavioral validate]
```

1. **Syntax validate** — the engine's `DQLSyntaxValidator` posts the DQL to the Grail language service's `POST /platform/storage/query/v1/query:verify` endpoint for parser-level validation. No data executes.
2. **Tenant validate** — `DTEnvironmentRegistry` checks that referenced metrics, entities, buckets, and OpenPipeline enrichment attributes actually exist in the target tenant.
3. **Behavioral validate** — run both NRQL (against NR) and translated DQL (against DT) over the same time window and compare result shape and magnitudes.

Auto-fix

When syntax validation fails, the engine attempts safe corrections before reporting failure:

- Operator normalization (`>` becomes `>=` if NRQL was inclusive)
- Duration literal correction (`1m` → `1m` ensures `m` suffix lowercase)
- Field name normalization (camelCase ↔ snake_case for known mappings)

Applied fixes are returned in `result.fixes[]` so reviewers can see what changed.

Summary

NRQL→DQL translation is a compiler problem, not a string-replace problem. The open-source `nrql-engine` covers 292 patterns with confidence scoring; LOW-confidence queries are flagged for manual review rather than silently emitted as wrong DQL. Pair the translator with the validation loop, treat MEDIUM confidence as a review obligation, and keep an explicit gap-analysis for the genuinely unsupported constructs.

Continue to **NRLC-03 Dashboard Migration** to see how translated queries flow into Dynatrace dashboard artifacts.
